Chemistry 3A – Spring 2026 – WITH FEEDBACK **Version A**

Sections 19320, 19321

Midterm Examination #2

Select the **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt

1. Of the following intermolecular forces dipole-dipole (D-D), hydrogen bonding (H-B), London Dispersion Forces (LDF), which choice shows the order of strength of interaction? (LDF > H-B means LDF is stronger than H-B)

a) LDF > H-B > D-D b) H-B > D-D > LDF c) H-B > LDF > D-D d) D-D > H-B > LDF

Of the intermolecular forces discussed, the instantaneous or temporary induced dipole is an interaction between NON-POLAR molecules that is considered the weakest compared to molecules with a permanent dipole nature (two polar covalent or one polar and one nonpolar covalent. The strongest is hydrogen bonding, so the strength should be H-B > D-D > LDF

2. On the Periodic Table is a number with each element. For example, for Hydrogen (H), that number is "1.008" or close to that value. What does that number represent?

a) electronegativity b) grams c) density d) molar mass

The Periodic Table has the name, symbol, and the atomic number (Z value). The first element has name "Hydrogen", symbol "H" and atomic number 1. It often has the number "1.008" or perhaps "1.0079". All chemists understand that this is the weighted average of the isotopes of the atoms of the element. This number also represents the number of grams for one mole of atoms, and so the units are correctly stated as "grams / mole" or with proper abbreviation, "g/mol". The units "g/mol" by definition indicate the **molar mass**. Electronegativity can be indicated on a Periodic Table, but that is rare for most table presentations, and besides, the electronegativity of hydrogen is a value of about 2.1 (which itself is an arbitrary scale although backed up by experimental evidence. The choice **grams** cannot be true because these are not correct units and indicate a fundamental misunderstanding of what the table is for, what it represents.

3. Which of the following represents a change in enthalpy in a substance that is being is being boiled?

a) ΔH_{vap} b) ΔH_{sub} c) ΔH_{fus} d) ΔH_{melt}

ΔH_{vap} is the "heat of **vap**orization". Vaporization is the process of matter in a liquid state going to a gaseous state. The term "boiling" has been used to describe that same process: of heat energy being added to a liquid that reaches a temperature called its boiling point. It is at this temperature that when heat energy is added to the liquid, it goes to ***boiling*** the liquid into gas/vapor accompanied with no temperature change into the matter has fully gone to gas/vapor.

ΔH_{sub} is the "heat of sublimation", which is the energy related to converting matter from a solid state into a gaseous state, with no intermediate liquid state. This can happen with compounds like carbon dioxide (CO2), when it transitions from dry ice (solid CO2) to the typical standard gaseous state at room temperature.

ΔH_{fus} is the "heat of fusion" is the energy related to converting matter from the solid state to the liquid state, which occurs at a constant temperature known as the **melting point.**

ΔH_{melt} is the "heat of melting". It is not really a proper symbol or naming as it should be ΔH_{fus}, but this symbolism is understood to be used by those learning chemistry. ΔH_{fus} should always be used by chemists.

All (heat) energy units should be in Joules (J) or calories (cal) in chemistry.

4. Which of these types of matter will have a definite volume but not a definite shape?

a) gas b) liquid c) solid d) London Dispersion Force

For the states of matter gas, liquid, solid, the physical properties discussion were about whether these states had a definite volume and shape. For solids, they have a definite shape and volume. For liquids they have a definite volume but their shape was flexible, could be changed. For gases, they do not hold a definite shape or volume on their own (a container is something external that can be said to give the matter shape or volume intrinsically). London Dispersion Force is a distractor choice easily removed with simple logic.

5. What is the mass percentage of water in the hydrate $CuSO_4 \cdot 5\ H_2O$?

a) 7.23% b) 36.08% c) 51.16% d) 63.92%

There are two steps. But first is to get the proper count of ALL atoms in a compound/molecule:

One Cu (copper). One S (sulfur). Now for hydrogen and oxygen. Rule: the coefficient (if not present, then the coefficient is "1" for calculation purposes) multiplied with the subscript. Start with oxygen (O): it is present is two places: (a) in copper sulfate and (b) in the hydrate. There are 4 atoms in sulfate, 5 x 1 atoms in water, total of 9 atoms. For hydrogen, there are 5 x 2 = 10 H atoms. So the count is complete. Now it's time for addition using the molar mass for each atom from the Yellow Sheet Periodic Table: 1 x 63.55 (Cu) + 1 x 32.06 (S) + 9 x 15.999 (O) + 10 x 1.008 (H) = 249.681 amu (or g/mol). Now just do the water alone: 5 x 1 (O) x 15.999 + 5 x 2 (H) x 1.008 = 90.075 amu (or g/mol). Now compute the percentage: 90.075 / 249.681 x 100% = 36.08%

6. Assume specific heat capacity of cadmium (Cd) is 0.250 J/(g °C). A 10.0 g piece of cadmium metal is heated from 20°C to 50°C. How many joules of heat energy were transferred to the metal?

a) 400 J b) 225 J c) 75.0 J d) 0.635 J

We apply that equation $q = m \times c \times \Delta T$, where q symbolizes energy in the form of heat (energy takes many forms in physics/chemistry, including "work") and its units are either in Joules (J) or calories (cal) for chemists; m is the mass of the object that is being gaining (heating) or losing (cooling) energy; ΔT is the change in the temperature of the mass/object, and it must be calculated as FINAL temperature minus the INITIAL temperature ($T_{final} - T_{initial}$), with units of °C (Celsius) or K (Kelvin); and the symbol c is the **specific heat capacity**, which is experimentally determined for any compound/molecule as matter in a particular physical state (gas, liquid, solid). The specific heat capacity is a physical property that measures how much thermal energy is required to raise the temperature of exactly 1 gram of a substance by 1 °C or 1 K; the units are "J / (g °C)" or "cal /(g °C)". (It can also be "J / (g K)" or "cal / (g K)" but rarely.) Think of it as a measure of "thermal inertia"—it tells you how much a substance "resists" changing its temperature when heat is added or removed.

Those 4 values (m = 10.0 g , c =0.250 J / (g °C), T_{final} =50°C, $T_{initial}$ = 20°C) are the inputs to getting the q value. 10.0 g × 0.250 J / (g °C) × (50°C – 20°C) = 75.0, with 3 significant digits required and found

among the choices. Note that all units in the expression cancel except Joules (J), which is what is expected to calculate a (heat) energy value.

7. What is the best choice for what is TRUE about the characteristics of a Type II ion?

a) it is a polyatomic anion composed of nonmetal elements
b) it is an element that will ionize only to one charge state, such as +1 or +2 or +3 or more, but only one of those
c) the element's name in a compound requires the use of a roman numeral (the Stock system)
d) it only applies to Group 2 elements, which is why it is called a Type II ion

With these questions, the strategy is always to evaluate each choice for being true or false. The objective is to find the ONE true statement in a choice. Each choice must be whether the statement is about a Type II ion. Choice (a) is false since Type II ions are positively charged (cations) and not composed of nonmetals, whether monatomic or polyatomic. Choice (b) says that the element has only one ionic charge (like sodium can only be Na^{+} or calcium is Ca^{2+}), and this is what a Type I metal cation is. Choice (d) is a common mistake in which Group 2 elements typically have a fixed 2+ charge (like Mg^{2+} or Ca^{2+}) and so it is incorrectly believed that Type II cations have a 2+ charge; Type II refers completely to something else. Choice (c) is the correct choice and TRUE statement: one should ALWAYS think that if an ion is Type II, it requires the roman numeral (Stock system) for naming the metal cation; they go hand in hand.

8. In drawing a Lewis structure for the nitrate $(NO_3)^{-}$ ion, you have placed the octet around the three oxygen (O) atoms, but you see the central nitrogen (N) atom has only six electrons (3 bonding pairs) to the three oxygen atoms. You have no electrons remaining to add in your inventory. What step do you need to do?

a) Nothing: your Lewis structure is ready and complete
b) Take a lone pair from one of the oxygen (O) atoms and use it to create a double bond with the central nitrogen (N)
c) Add a hydrogen atom to the molecule
d) Use Avogadro's Number at an earlier step

This is the review of the steps in building Lewis structures!
(i) Count the valence electrons of all atoms in the molecule PLUS the electrons in the superscript if it is anion (negatively charged) OR MINUS the electrons in the superscript if the ion is a cation (positively charged). Be very careful in this step. This is your starting inventory of electrons

(ii) Arrange the atoms with the approximately correctly geometry, taking care to determine the CENTRAL atom and all the TERMINAL atoms

(iii) Now put single bonds between the atoms as they exist, usually between the CENTRAL atom and the TERMINAL atom. Although a bond can be shown with a pair of electrons (as dots) in doing Lewis structures, use lines and not dot pairs to show bonds. It makes it less confusing. Now count each bond and multiply by 2, and subtract that total from your electron inventory

(iv) LONE PAIR PLACEMENT: if you have TERMINAL oxygen (O) atoms, put the proper number of lone (nonbonding) pairs around these atoms, taking care of the octet rule. This includes other atoms with lone pairs that may be terminal atoms, such as the halogens (F, Cl, Br, I). This is NOT to exclude other atoms too that may NOT be the CENTRAL atom, namely nonmetal and semi-metal atoms. Make sure to count all electrons used and subtract those from the inventory.

(v) LEFTOVER ELECTRONS: Got any electrons left? Those will usually be lone pairs around the CENTRAL atom. If you have an ODD number of electrons, make sure first that the molecule did not have a net charge. If the molecule is truly neutral (net zero charge), then put electrons as lone pairs and then one single (odd) electron in the central atom.

(vi) COMPLETING (FORCING) THE OCTET RULE (MAKING DOUBLE and/or TRIPLE BONDS): This is just a matter of making sure all atoms that should have an octet (hydrogen has a 'duet") have that octet. If necessary, a lone pair on an atom is moved into a bond (bonding pair) to form another bond or two, making a double or triple bond.

(vii): this is just the triple-check rule of reviewing steps (i) through (vi). FORMAL CHARGE ANALYSIS: look at each atom of the molecule: determine how many electrons in its valance shell are "owned" by the atom, which is all the lone pair electrons and one-half of the electrons in any bond it has with other atoms, and how many electrons it should have if it has a zero charge (the number of electrons it brought to the molecule). If the electrons "owned" is in excess of electrons for zero charge, the formal charge on the atom is negative, if the converse, the formal charge is positive. Now add up the formal charge on all atoms, and it should equal the net charge on the molecule. Use this formula:

Formal Charge = Valence e^- – Non-bonding e – ½ Bonding e^-

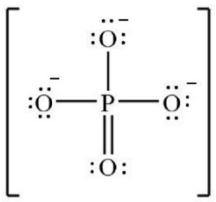
9. What is the molar mass of $CaSO_4$?

a) 88.14 g/mol b) 104.14 g/mol c) 136.14 g/mol d) 152.14 g/mol

This is simply using the Periodic Table to calculate a molar mass, which is something all students should be comfortable in doing now. Step 1 to determine the number of all the unique atoms in a compound/molecule. One Ca atom, one S atom, and four O atoms. For a more complex molecule/compound like subscripted polyatomic ions, say calcium nitrate, $Ca(NO_3)_2$, you multiply subscripts with subscripts: One Ca, 2 times one N (= two N atoms), and 2 times three O atoms (= 6 O atoms). With coefficients like a hydrate, then multiply the coefficient and the subscript. But most of the time, you must keep track of subscripts in any compound.

Ca = 1 x 40.08 g/mol + S = 1 x 32.06 g/mol + O = 4 x 15.999 g/mol = 136.14 g/mol when you round it off at 2 decimal places. Choice (c) was your best answer. If this calculation is at all confusing, CONTACT ME IMMEDIATELY!!

The Lewis structure is shown for PO_4^{3-} ion at right. **Next four questions** relate to it.



10. What are the total number of electrons (valence electrons of atoms and electrons that account for the charge) for the molecule?

a) 14 b) 18 c) 22 d) 26

Unbelievably, a correct choice was not offered to you. So the points on this are given to you freely. This is basically Step 1 in building a Lewis structure: count the valence electrons of each atom in the molecule and then either add or subtract to that value the net charge on the molecule.

The P atom brings 5 atoms electrons, and each O atom brings 6 valence electrons. You should know how to do that using the Periodic Table and what you have been told about what the valence shell electrons (versus core shell electrons) and how important they are in chemistry. That is a total of 29 electrons. Because the net charge on the molecule has been given to you in text (PO_4^{3-}) and visually with the negative signs on 3 of the 4 O atoms, indicating all formal charges on those 5 atoms, we

need to ADD three more electrons, making a total of 32 electrons. 32 should have been ONE of the choices, but was not. In almost every case, the total number of electrons in a stable molecule will be an even number of electrons (the odd-electron exceptions have been noted).

11. Each atom in the molecule should have an **octet** of electrons. How many electrons is that?

a) 4 b) 6 c) 8 d) 12

This is actually asking you if you know the definition of octet. The prefix "octo-" is Latin and "octa-" is Greek for "eight" (8), so this goes way back. Choice (c) is correct.

12. Which of the following statements is FALSE about the Lewis structure

a) One of the oxygen atoms has two nonbonding (lone) pairs of electrons
b) There are a total of 3 bonding pairs of electrons in the molecule
c) The molecule has a net charge of -3
d) The central atom phosphate (P) has no nonbonding (lone) pairs of electrons

This question requires a T-F evaluation of each choice testing your knowledge of Lewis structures in general, and point to the phosphate ion (PO_4^{3-}) in this question set. Choice (a) is TRUE because of the four terminal O atoms in phosphate, three have 3 lone pairs (and thus a formal charge of -1) and the one double-bonded terminal O atom does have two lone (nonbonding) electron pairs. Choice (c) is TRUE because phosphate ion clearly has a -3 net charge. Because the phosphorus (P) atom has no non-bonding (lone) pairs, choice (d) is TRUE. Finally phosphorus has 3 single bonds and 1 double bonds and actually has 10 (or 5 pairs) of electron involved in bonding, with 4 electron groups and tetrahedral geometry for electron groups and molecular.

13. Which of the following is a TRUE statement about the molecule

a) This is an excellent example of a **cation**
b) The central P atom has four (4) **electron groups**
c) There is -1 formal charge on all four O atoms
d) This molecular has a **trigonal planar** molecular geometry

Continuing with the phosphate ion, analyze each choice for truth. Choice (a) says it is an example of a cation, but since a cation is a **positively** charge ion by definition, and phosphate has a -3 charge, that is FALSE. The central P atom has four electron groups based on the definition of electron groups (which are either atoms or lone pairs) and phosphate has four atoms (oxygens) bonded to it, so this is a candidate for being the required TRUE choice. There is -1 formal charge on three of the four O atoms, and the fourth has a formal charge of zero, so choice (c) is FALSE. With four O atoms bonded to the central P atom, the molecular geometry is predictably tetrahedral and not trigonal planar. We go back to choice (b) as correct.

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14. Methane (CH_4) is a molecule with four terminal H atoms and no nonbonding (lone) pair of electrons on the central C atom. What is its electron group geometry?

a) bent b) trigonal planar c) trigonal pyramidal d) tetrahedral

Numerous times in lecture methane was given as an example of having tetrahedral electron group geometry and molecular geometry. NH_3 and H_2O were also given as examples, although all three have different molecular geometries. Review the lecture set 6.

15. Hydrogen (H) and fluorine (F) show a somewhat large difference in their electronegativity values. But not like sodium (Na) and chlorine (Cl). What kind of chemical bond are H and F likely to form as a molecule?

a) ionic b) metallic c) (nonpolar) covalent d) polar covalent

Chemical bonding between atoms is all about valence electrons. Ionic bonding is about one atom, a metal element, usually giving up all its valence electrons to another atom, a non-metal atom, wanting one or more electrons, both trying to stabilize their sets of electrons to look like a noble gas element electronic configuration. Metallic bonding is another type of bonding between neutral metal atoms who valence electrons are "loosely held" in a freely mobile "sea of electrons", making metals good electrical conductors. Bonding between the nonmetal elements, which form polyatomic molecules, some of which become ions, can be polar covalent or (nonpolar) covalent, depending on the differences between electronegativities between the two nonmetal atoms. Carbon (C) and hydrogen (H) have little differences in electronegativities, which means no one atom has a greater "pull" or attraction of the bonding pair(s) of electrons to itself (the definition of electronegativity). But certain nonmetal atoms, F > O > Cl > N > Br > I > S > Se, when bonded with hydrogen will create a strong polarity, with hydrogen having its electronic shell shield pulled away and enabling hydrogen bonding where possible. This type of bond will be polar covalent, which is choice (d).

16. A diamond is composed of pure carbon (C) and forms an extensive structure of tetrahedral geometry What class of (crystalline) solid does extensively hydrogen-bonded H_2O as ice as well as carbon dioxide (CO_2) as dry ice, and also diatomic element molecule iodine (I_2) form?

a) ionic b) covalent network c) metallic d) molecular

This tests the knowledge of classes of solids. A diamond is mentioned and this is a covalent network solid. But then solid H_2O (ice) and CO_2 (dry ice) and iodine (I_2) are exampled of molecular solids (choice (d)). Ionic solids would be crystals of a salt, and metallic solids examples are like gold bars/ingots. Molecular solids are molecules that use intermolecular forces in some way to come together to form a solid. Choice (d) is correct.

17. Water is at 100°C. What is its temperature on the Kelvin scale?

a) 0 K b) 100 K c) 298 K d) 373 K

This determines your ability to convert between Celsius and Kelvin temperature scales. 0°C = 273 K (I allow integers here). °C = K – 273. K = °C + 273. Simple algebra. Thus K = 100 + 273 = 373 K. Choice (d) again.

18. When atoms like rubidium and chlorine have the largest differences in electronegativity, what kind of bond will they form?

a) ionic b) metallic c) polar covalent d) covalent-ionic

Electronegativity reflects a property of natura/matter that some atoms have a stronger affinity for electrons than others. It's all about achieving a stability that looks like the configurations of the noble gas elements. Group 1 and 2 elements, like rubidium, will typically lose the one or two electrons in their valence shells and so their electronegativity values are very small compared to the Group 16 and 17 elements, like chlorine, which have significant electronegativity. Effectively the attraction for these electrons results in these elements becoming ions (ionized), and the resulting ion formation then creates a bonding that is electrostatic (based on electric charge formation), those forces being very strong.

19. If you have a Ba^{2+} ion and a SO_4^{2-} ion, what compound will it form as you apply the criss-cross method?

a) Ba_2SO_4 b) $Ba_2(SO_4)_2$ c) $BaSO_4$ d) $Ba(SO_4)_2$

The criss-cross method is used to construct the chemical formula of a compound, typically an ionic compound. The ionic charge value in the superscript of the ion is used to write the chemical formula by placing that charge value (taking away the plus and minus signs) as the subscript of the other atom in the compound. Consider sodium cation (Na^+) and phosphate anion (PO_4^{3-}) to form sodium phosphate. Following the criss-cross, one takes the implied "1" of the "1+" of sodium and puts that as a subscript in the phosphate. Because phosphate is a polyatomic, first put parentheses around the phosphate then put the subscript "1". Now take the "3" of "3–" in phosphate (PO_4) superscript, and make it the subscript of sodium (Na). So the result so far is "$Na_3(PO_4)_1$". Since a subscripted "1" Is not necessary, the current result is "$Na_3(PO_4)$", and this further becomes "Na_3PO_4" since a single polyatomic group does not need the parenthesis. The final formula is obtained.

Now to the question: apply the criss-cross to get the formula $Ba_2(SO_4)_2$. But since both subscripts are further reducible (divisible by 2), the result is Ba(SO4), and since there is only one in number of the polyatomic, then drop the parentheses: $BaSO_4$

20. How many moles of a compound is there if the number of molecules or atoms or particles of that compound is five (5) times Avogadro's Number?

a) 0.2 mol b) 1 mol c) 5 mol d) 20 mol

Tests the fundamental understanding of the mole (unit = "mol") as a number or count of particles, and that that number or count for one mole is Avogadro's Number (NA), which is a constant determined to be 6.022×10^{23}. Since the number is in the question is five times NA, and 6.022×10^{23} = 1 mol, then (5 x 6.022×10^{23} particles/atoms/molecules) x (1 mol / 6.023×10^{23} particles/atoms/molecules) = 5 mol. Choice (c)

21. If a compound has a molecular formula of $C_4H_{10}N_2$, what would be the empirical formula for it?

a) $C_4H_{10}N_2$ b) $C_8H_{20}N_4$ c) C_2H_5N d) CHN

In a procedure called "elemental analysis", this determines the mass of all component elements of a compound/substance, and then divides each mass of the element by the element's molar mass, coming up with the actual moles of the element. It then divides all the mole determinations of each element by the value of the element with the lowest number of moles. The element with the lowest number of moles will have a mole value of "1.xxx" and all the other elements should have values greater than "1". A final adjustment of these values is done to have each element have an integral (whole number) value since chemical formula are composed of whole atoms and not fractions of atoms. The result should be the compound's **empirical formula**. It will not be the **molecular formula**, however, which will be a multiple of the empirical formula. In this case, it will be necessary to divide by 2 the molecular formula

So it is said that if you have the empirical formula and you don't know the molecular formula of the compound, that you need to get the molar mass of the compound to determine the molecular formula. Great! So how do you get the molar mass? There are more methods that have to be done. The gold standard is to do mass spectrometry. Other lower cost methods would be to add compound to a liquid like water and see how it depresses the freezing point of water or elevates its boiling point.

22. What kind of **intermolecular force** involves a hydrogen atom bonded to a very electronegative atom and causing a strong attraction to a lone (nonbonding) pair of another electronegative atom?

a) London dispersion forces
b) hydrogen bonding
c) covalent-ionic attraction
d) dipole-dipole interaction

There were 3 classes of IMF discussed in lecture. The description in the question is close to the very definition of hydrogen bonding, making choice (b) the correct answer. Choice (c) is a distractor as it is a nonsense term never discussed. London dispersion forces and dipole-dipole interaction were the other two discussed, but are not described here.

23. What is the correct name for **$FeCl_3$**, noting that Fe is a Type II metal cation?

a) tin(IV) chloride b) iron(II) chloride c) iron(III) chloride d) iron chlorine

The question could have not added the "noting that Fe is a Type II metal cation" part, but this was offered as a helpful clue that the name would include a roman numeral. Three of the choices did in fact offer a roman numeral in the name. The part not helped is in determining which form the iron cation is. It is necessary to know that chlorine forms only one ion, the –1 chloride anion. And since there are three chlorine atoms in the compound, that requires that iron have a 3+ charge to balance to form the neutral compound. Iron does have a +2 and +3 form (which is why it is Type II) of its ion. But since the determination is +3, this means the roman numeral 3 "III" must be used in the name, meaning choice (c) is correct. Choice (a) requires a chemical formula of $SnCl_4$ which is not given. Choice (b) would require a formula of $FeCl_2$, and choice is some incorrectly named compound—should be "chloride"—and it is missing information on the roman numeral

24. Which molecular formula correctly shows the compound **dinitrogen tetroxide**?

a) NO b) N_4O_2 c) N_2O_2 d) N_2O_4

Compound naming particularly of the nonmetals with the prefixes 1-10 (mono-, di-, tri-, tetra-, etc) was explored in a laboratory session we did. Correct choice is (d).

25. There are 6.022×10^{23} atoms of element potassium (K). How many grams of potassium are there?

a) 6.022×10^{23} g b) 6.65 g c) 40.08 g d) 241.3 g

This question would have been a good one if one of the choices given had the answer, which it did not. The question was borrowed from the 2nd midterm of the previous semester, but the choices had not been updated to include the correct choice of "39.10 g", so this question becomes a free one.

26. 1 mol of sodium (Na) and 1 mole of chlorine (Cl) are equal numbers of Na and K atoms

a) true b) false

The question should have been written "1 mol of sodium (Na) and 1 mole of chlorine (Cl) are equal numbers of Na and Cl atoms" OR "1 mol of sodium (Na) and 1 mole of potassium (K) are equal numbers of Na and K atoms" but because of the confusion, the answer could have been true or false. Written correctly the only answer would have been TRUE, because a MOLE is a count of particles—in fact, Avogadro's Number of particles (6.22×10^{23}), and one mole of any substance is equal to a mole of another substance.

27. Chemical potential energy is stored energy in a chemical bonds

a) true b) false

Kinetic energy is the energy of matter that changes position, or rather, has a velocity. Potential energy is the energy stored in matter than often has a force applied to it. For instance, an object on the 10th floor of a building has an energy that will "show itself" when that object is released and allowed to fall; the force applied to it is the force of gravity.

In the same respect, energy is stored in chemical bonds too. In combustion, when oxygen (O_2) reacts with a combustible compound like methane (CH_4), the energy is breaking the C-H bonds is larger than the energy in forming the bonds that produce carbon dioxide (CO_2) and water (H_2O)

28. Nonbonding (lone) pairs of electrons affecting molecular geometry is explained by Valence Shell Electron Pair Repulsion (VSEPR) theory
a) true b) false

The main concepts involving VSEPR were the effect of lone pairs of electrons creating significant repulsion in the central atom of the molecule to affect the geometry of a molecule. There were two aspects of that geometry, one being those by the electron groups of the central atom (bonded atoms plus lone pairs) contributing to "electron group geometry" and the atoms-only groups that affected "molecular geometry". The statement is therefore FALSE since the lone pairs affect the one geometry (EGG) and not the other (MG)

29. The freezing of water to ice is an endothermic process
a) true b) false

Recall that we have a system and surroundings. Let's focus only on the system which is the water. When energy in form of heat goes INTO the system, it is ENDOthermic. When energy in form of heat is goes OUT OF the system, it is EXOthermic. (think "ENDO" = INTO system, "EXO" = "exiting" or OUT OF system. Now when you melt ice or boil water, you have to put energy INTO system, hence ENDOthermic. When you do the opposite, namely freeze water or condense system, you have to take energy as heat OUT OF the system, therefor EXOthermic. You have to think of energy, in what form it is, and in which direction the energy is going/"flowing". But even I get confused on that when logic gets tangled.

30. Calcium is a Group 13 element
a) true b) false

This is just looking at the Periodic Table and seeing if you understand its structure. Statement is FALSE.

31. Beryllium has a molar mass of 9.012 g/mol
a) true b) false

Like Question 30, how well do you understand the Periodic Table in the Yellow Sheet. Answer is TRUE.

32. There are more molecules in 0.50 mol of CO_2 than in 0.50 mol of SO_2
a) true b) false

A mole represents a count or number of particles, atoms, molecules or formula units. It is NOT a mass, measured in grams or atomic mass units. Clearly the mass of CO_2 and SO_2 are different (in amus, or in grams). 0.50 mol of any substance will have the same number/count of particles, atoms, molecules, or formula units. Answer FALSE.

33. London dispersion forces are the weakest of intermolecular forces where dipole-dipole interactions and hydrogen bonding are the other forces.

a) true b) false

This may be the 3rd question related to intermolecular forces, with the added comparison hint of dipole-dipole interactions and hydrogen bonding. Answer TRUE

34. Condensation is the process of a solid turning directly into a gas.

a) true b) false

Sublimation is the process of a solid as matter being transformed into a gas. Condensation is the process of a gas becoming a liquid.

35. Sulfur dioxide (SO_2) is an ionic compound.

a) true b) false

In the same respect, energy is stored in chemical bonds too. In combustion, when oxygen (O2) reacts with a combustible compound like methane (CH4), the energy is breaking the C-H bonds is larger than the energy in forming the bonds that produce carbon dioxide (CO2) and water (H2O)